

3.2.8 Responsible design

Environmental issues

Content	Potential links to maths and science
Students should be aware of, and able to discuss, the importance environmental issues in design and manufacture, including: • the responsibilities of designers and manufacturers in ensuring products are made from sustainable materials and components • the environmental impact of packaging of products, eg the use of excessive packaging and plastics.	

Conservation of energy and resources

Content	Potential links to maths and science
Students should be aware of, and able to discuss, the concept of a circular economy, including: • how products are designed to conserve energy, materials and components • the design of products for minimum impact on the environment including raw material extraction, consumption, ease of repair, maintenance and end of life • sustainable manufacturing including the use of alternative energy and methods to minimise waste • the impact of waste, surplus and by-products created in the process of manufacture including reuse of material off-cuts, chemicals, heat and water • cost implications of dealing with waste • the impact of global manufacturing on product miles.	